

Mars image anomaly detection

NSSC 2026 | My preprocessing experiments and final analysis

I trained an autoencoder from scratch and used Isolation Forest on its latent vector. The first candidate list was dominated by unusually bright images and changed across repeated runs. I added controlled brightness and footprint experiments to investigate that weakness.

My decision

I retain v3 as the screening reference. v4 and v5 reduce a specified brightness nuisance, but their minimum forest-seed flag overlaps fall from 0.667 for v3 to 0.467 and 0.227. All five leading v4 examples contain conspicuous black strips or borders, and the v5 exact-zero mask leaves near-black edge fragments. That evidence does not support replacing the reference with a more reliable geological detector. v3 itself remains brightness-confounded and unstable across independent training runs; I do not present its candidates as confirmed anomalies.

Reference run: v3. The calibrated boundary is 0.542471; 17 of 10,422 crops exceed it. The 95% source-bootstrap threshold interval is [0.535230, 0.550470].

What this establishes. At fixed model, forest and threshold, v4 reduces the 95th-percentile absolute score response to $0.8 \cdot x + 0.1$ from 0.033935 to 0.002902 (91.4%). v5 reduces it by 99.87%. This establishes the intended illumination invariance on 512 fixed validation crops. No sampled flags flip in any run, and the transformation is deliberately affine; this does not establish better anomaly recall. The project now has stronger controlled evidence and a readable, reproducible notebook; a detector-accuracy improvement is not demonstrated.

What it does not establish. Hidden anomaly labels are unavailable. I cannot claim precision, recall, F1 or a verified false-positive rate. Candidate appearance and reproducibility are not ground truth.

Run	Input	Val. MSE	Val. SSIM	Flags
v1	raw	0.002694	0.6611	20
v2	raw	0.002740	0.6717	9
v3	raw	0.002599	0.6738	17
v4	contrast	0.007941	0.4975	24
v5	footprint	0.008554	0.4712	42
improved_seed	contrast	0.007757	0.5024	11
improved_split	contrast	0.006724	0.5349	10

Metrics use each run's best validation-objective checkpoint. Different preprocessing changes the reconstruction target, and the split repeat uses different validation images. Their MSE/SSIM are not interchangeable quality scores. All runs use random initialization.

The notebook is prepared for Colab/Kaggle. These saved experiments ran on a local RTX 3050; no cloud execution is invented.

Data and the preprocessing comparison

I decoded all 10,422 supplied crops, verified 227 by 227 grayscale inputs, checked metadata joins and found no exact decoded-pixel duplicates. There are 172 source observations. Near-duplicate or geographically overlapping observations remain possible.

Main split	Sources	Crops
Training	120	7,101
Validation	25	1,666
Calibration	27	1,655

I keep each source in one split. v4 changes only preprocessing: subtract the crop mean, divide by standard deviation, map with $0.5 + 0.15 \cdot z$, then clip to $[0,1]$. Standard deviation is floored at $1/255$. v5 adds exclusion of exact-zero pixels connected to the image border when estimating the transform, then fills them with 0.5. Disconnected interior zeros remain content.

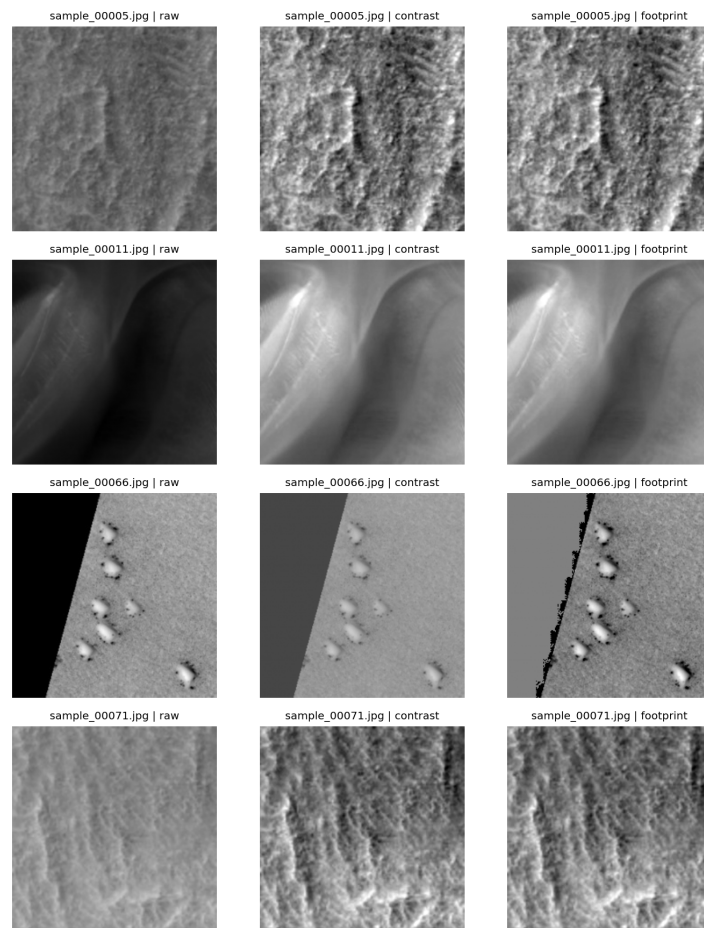


Figure 1. Fixed validation examples: raw, contrast-standardized and footprint-treated inputs. The mask is a heuristic, not an organizer-supplied validity mask.

This intentionally suppresses global brightness and can remove meaningful albedo information. Clipping can suppress extreme contrast; normalization can amplify noise; a connected shadow can be mistaken for an image footprint.

Phase 1 Architecture, loss and learning curves

Five stride-two convolution blocks use channels 16, 32, 64, 96 and 128, with GroupNorm and SiLU. Spatial sizes are 114, 57, 29, 15 and 8. A linear layer creates a fixed 128- or 256-dimensional vector. The decoder expands the vector and uses resize-convolution to reconstruct exactly 227 by 227. There is no bypass connection and no pretrained feature extractor.

v1 uses MSE. v2-v5 use $0.9 \text{ MSE} + 0.1 (1 - \text{SSIM})$, with an 11 by 11 Gaussian SSIM window, sigma 1.5 and data range 1. AdamW uses learning rate 0.0003, weight decay 0.0001 and batch size 16. The planned comparison gives each run 15 epochs, with best validation checkpoint selection.

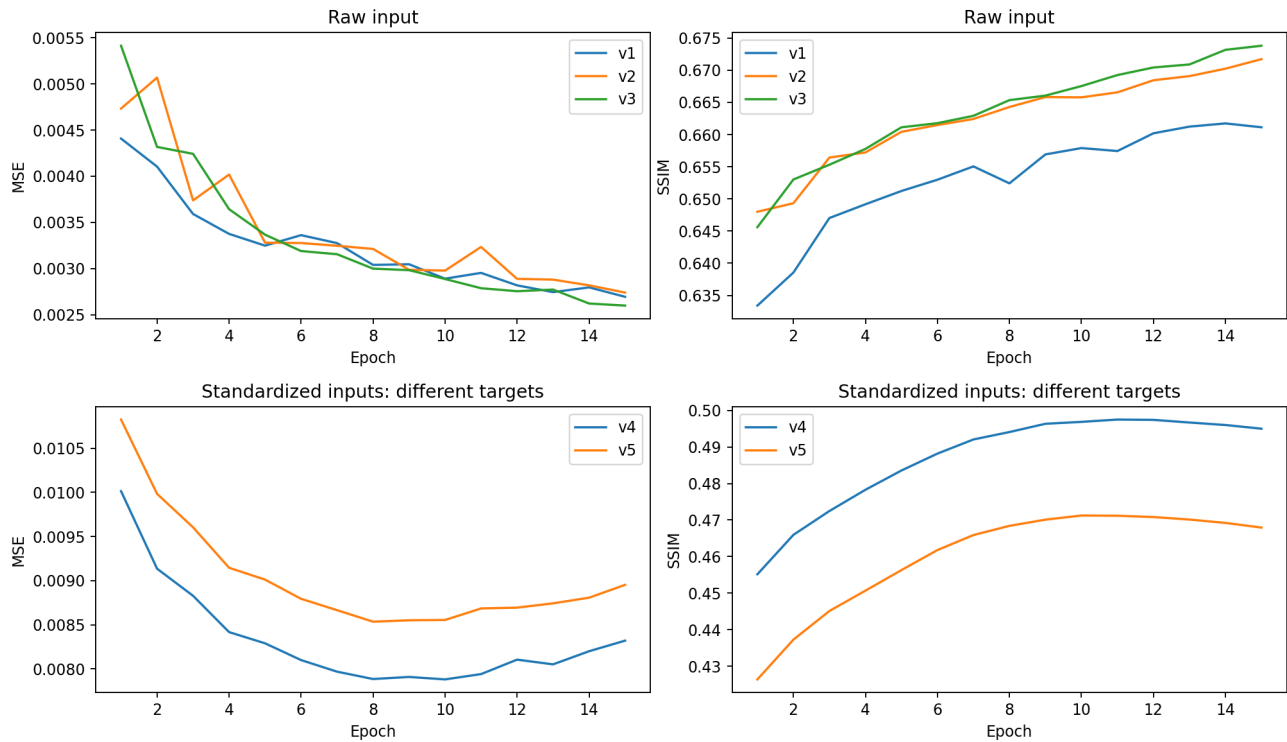


Figure 2. Raw-input and standardized-input learning curves are separated because their reconstruction targets differ.

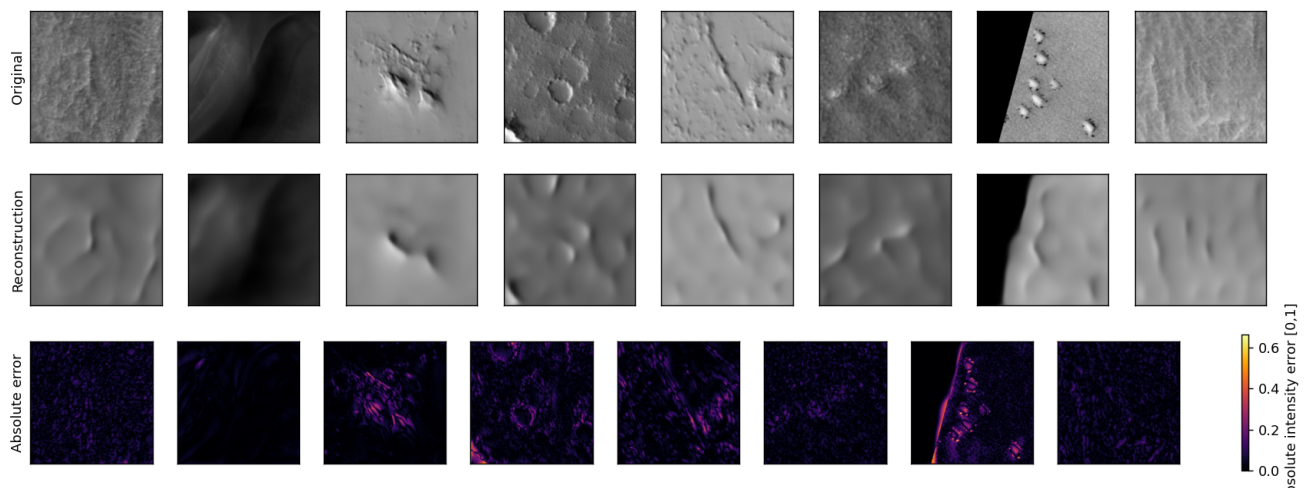


Figure 3. Fixed model inputs, latest reconstructions and error. Scoring uses the best checkpoint. The reference reconstructs raw inputs; standardized ablations label their different input domain explicitly.

Phase 2 Scoring and statistical calibration

I fit 2,000 Isolation Forest trees, max_samples 256, on training latents only. Novelty is negative score_samples: higher means more unusual. Metadata and reconstruction errors are not forest features, and the library prediction offset does not determine the flags.

The original adjusted-boxplot threshold for v1 was 0.87484, above its maximum score 0.55648. Its multimodal histogram motivated a distribution-model revision; I retained the zero-flag result. I did not lower a threshold merely to produce five images.

For every run, I fit one through four Gaussian mixture components on calibration scores, select minimum BIC, and define $T = \max_j(\mu_j + 3 \sigma_j)$. All fitted populations contribute to the envelope. No assumed anomaly count or contamination fraction determines the result.

The selected run uses 4 components, with fitted tail mass 0.000306. This is a fitted-model description, not a real false-alarm guarantee. Correlated crops make BIC heuristic, and unknown contamination can distort the model.

The component count reaches the search limit of four, which is a model-selection limitation.

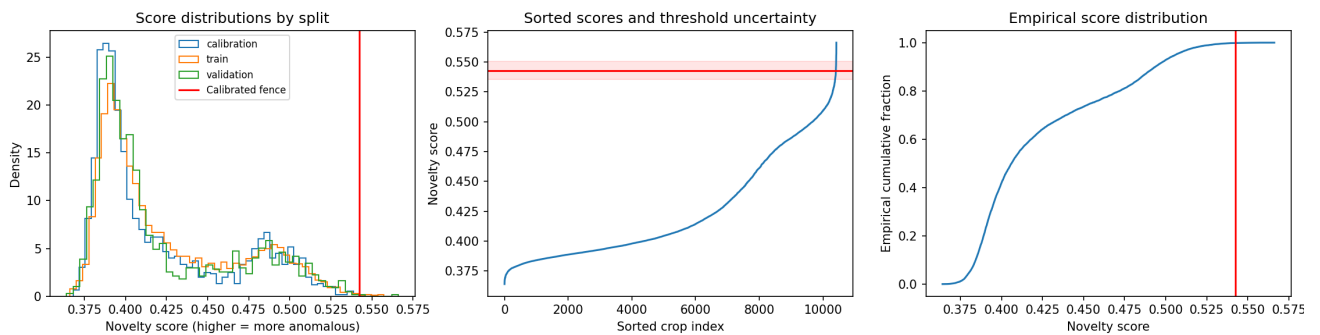


Figure 4. Split score distributions, sorted novelty scores and the calibrated threshold with source-bootstrap uncertainty.

I resample whole calibration sources and refit component selection in 100 bootstrap repetitions. The validation CDF maximum distance is 0.0597; I treat it descriptively and do not report an IID KS p-value. Conditional threshold flag frequency is not a probability of a genuine anomaly.

Did the improvement survive the checks?

For v4, changing initialization gives flag-set Jaccard 0.207 and full-ranking correlation 0.867; changing the source split gives 0.360 and 0.866. The three contrast runs flag 24, 11 and 10 crops, with 4 in their intersection. On 1,253 crops held out from both source partitions, rank correlation is 0.897. For raw v3, the corresponding initialization/split Jaccards remain 0.150/0.294, with zero crops flagged in all three raw runs. These are limited sensitivity checks, not detection accuracy. Initialization repeats change both encoder and forest seeds; separate forest-only repeats isolate detector randomness.

Comparison	Rank correlation	Flag-set Jaccard
v3 / robust_seed	0.949	0.150
v3 / robust_split	0.957	0.294
v4 / improved_seed	0.867	0.207
v4 / improved_split	0.866	0.360

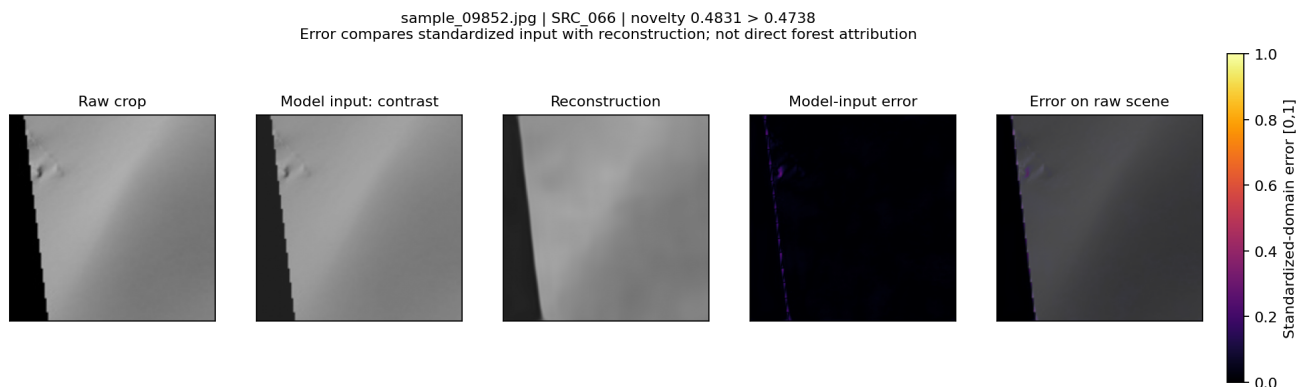
Jaccard is intersection divided by union of flagged sets. It can be low even when global ranking correlation is high. Runs with different splits also report common-held-out comparisons in the CSV. Neither metric measures anomaly accuracy.

Run	Median score change	95th percentile change	Flag flips / 512
v3	0.007651	0.033935	0
v4	0.000000	0.002902	0
v5	0.000000	0.000044	0

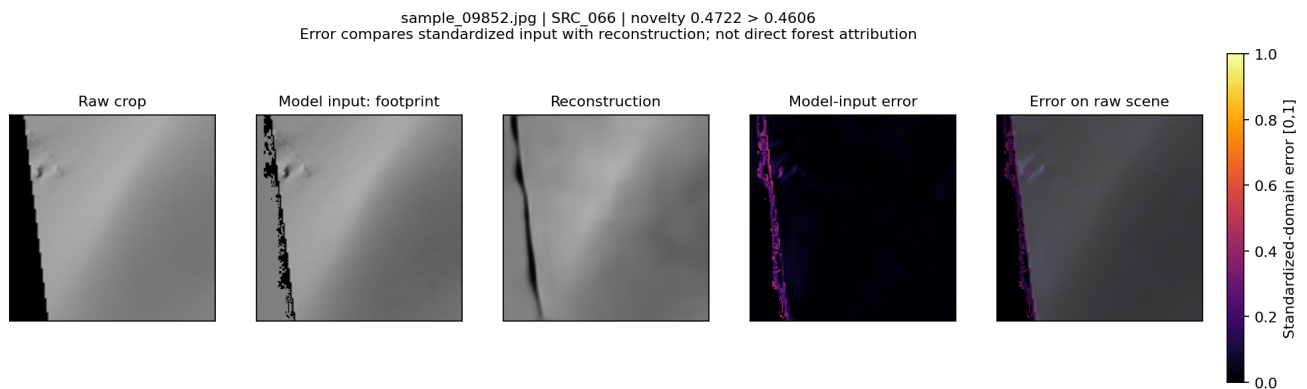
I hold each model, forest and threshold fixed, then apply $0.8 \cdot x + 0.1$ outside its original inferred black border on a fixed validation sample. This tests a specific photometric nuisance while preserving the footprint, not correctness on labelled anomalies.

Why better brightness invariance is insufficient

All five leading v4 candidates have conspicuous black strips or borders. Their brightness distribution is less extreme, but acquisition support remains a competing explanation. These examples were inspected after statistical selection; they were not used to lower the boundary or select a desired count.



v4: the same validation crop after the independently calibrated selection. The black border remains in the contrast-standardized input. Error belongs to the standardized input domain.



v5: the same validation crop after the independently calibrated selection. The footprint treatment leaves near-black edge fragments. Error belongs to the standardized input domain.

The controlled perturbation favors the invariance designed into normalization. It establishes that mechanism, but cannot establish preservation of real albedo anomalies. I retain v3 as the screening reference and preserve both negative ablations for the engineering journal.

Phase 1.3 Latent-space diagnostics

The reference vector has covariance effective rank 31.53 out of 256. It summarizes the variance spectrum, not a number of terrain classes.

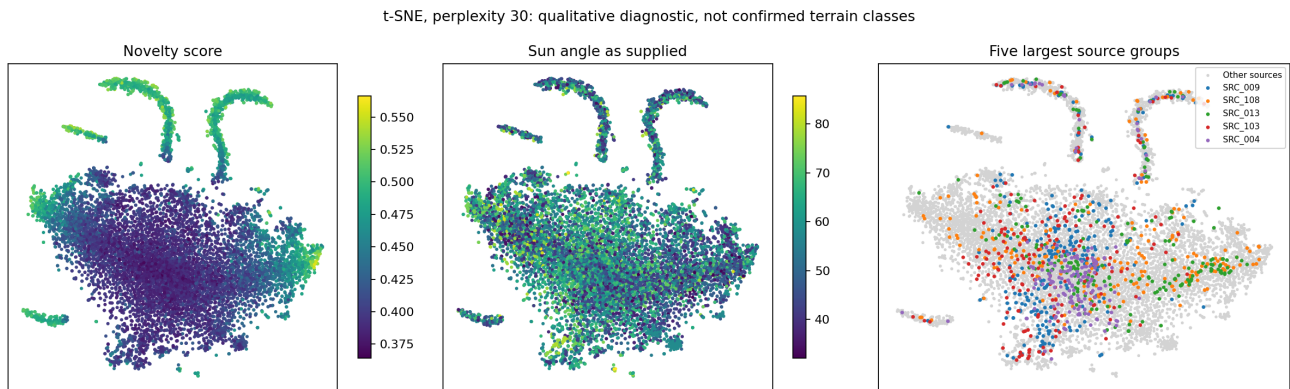


Figure 5. t-SNE at perplexity 30; novelty, supplied sun angle and five large source groups. PCA and standardization are for plotting only. The forest uses the full original latent.

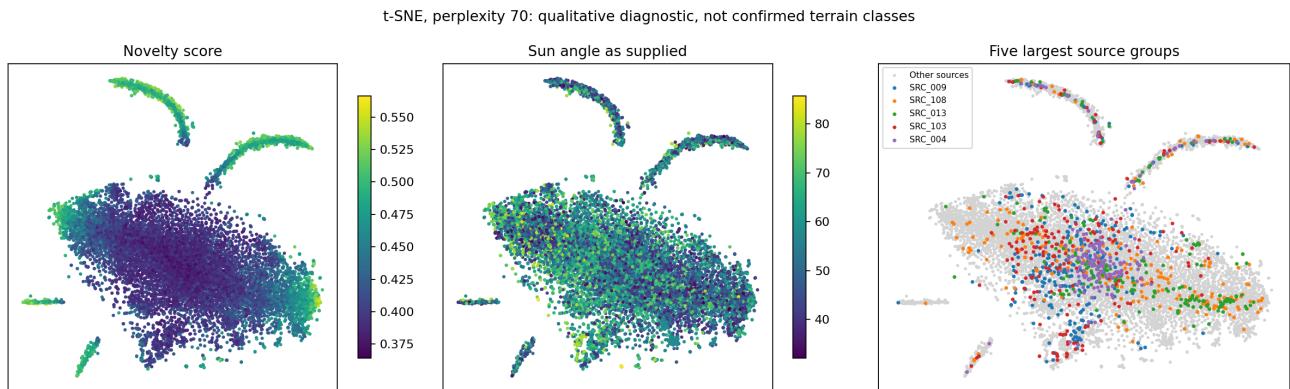


Figure 6. t-SNE at perplexity 70; novelty, supplied sun angle and five large source groups. PCA and standardization are for plotting only. The forest uses the full original latent.

The v3 latent projection is retained from the first-stage analysis. Two perplexities show broad structure, but these branches are not labelled terrain classes. Source and acquisition effects remain plausible; the between-source variance fraction is about 0.232.

Phase 2.3 Metadata and candidate context

I join only through the supplied crop-to-source index. Latitude has 24 distinct values, longitude only 0 and 180. Resolution units and sun-angle semantics are unspecified. These fields do not justify precise localization, physical dimensions or causal claims.

season	Crops	Flags	Crop rate	Equal-source rate
N-autumn	2181	6	0.275%	0.292%
N-spring	4190	2	0.048%	0.364%
N-summer	3030	8	0.264%	0.581%
N-winter	1021	1	0.098%	0.073%

resolution	Crops	Flags	Crop rate	Equal-source rate
0.25	5326	3	0.056%	0.215%
0.5	4959	13	0.262%	0.673%
1.0	137	1	0.730%	0.730%

The retained v3 flags 17 crops from eight sources. Its five selected examples all exceed the 99.7th brightness percentile, and three share SRC_154. For v4, median selected brightness falls to the 47.51st percentile and the largest source contributes 12.5% of flags, but the flagged median zero fraction is 13.36%. Source concentration and brightness improvement therefore do not remove acquisition concerns.

I keep metadata out of the novelty score and do not claim the optional fusion credit. Equal-source summaries reduce domination by large source observations, but they are descriptive.

Phase 3 Candidate 1

sample_07899.jpg | SRC_128 | validation

Novelty 0.566127 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

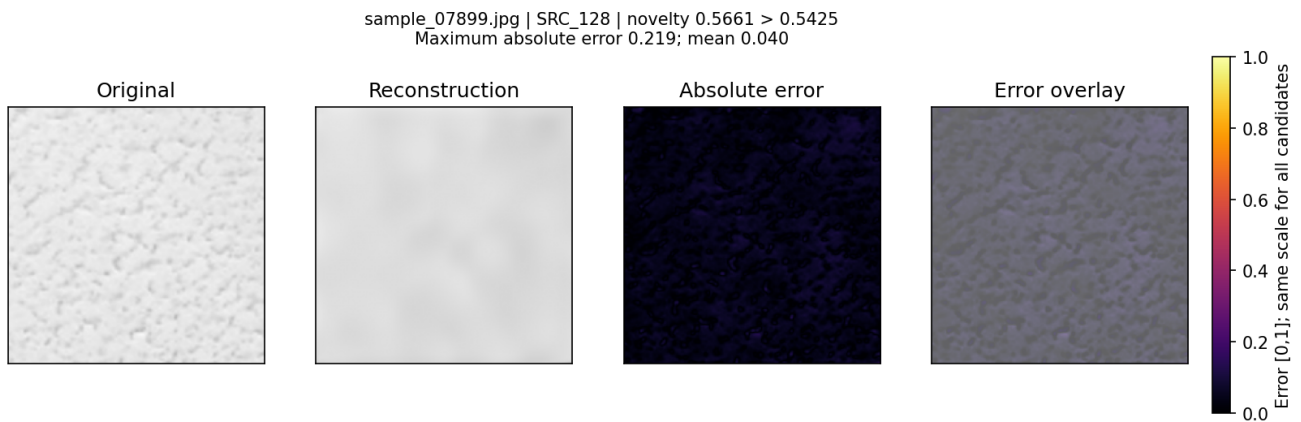


Figure 7. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. A very bright crop with irregular, fine dark mottling across most of the image. The reconstruction retains broad brightness but loses the small-scale pattern. Error is distributed through that texture rather than concentrated on a single foreign object.

Possible physical explanation. A bright surface mantle, potentially seasonal frost over textured polar terrain, is a plausible ordinary explanation. The supplied latitude of 90 and N-spring category are consistent with a polar-seasonal interpretation, if those coarse metadata values are representative.

Alternative and limit. Bright dust, exposure or contrast processing could also produce this appearance. A grayscale crop cannot establish ice composition, exact location or artificial insertion. The seasonal interpretation is a hypothesis, not a catalog identification.

Supplied metadata: latitude 90.0, longitude 0.0, sun angle 56.634817, season N-spring, resolution 0.5.

Process background, not crop identification: https://www.uahirise.org/PSP_007712_2670.

Phase 3 Candidate 2

sample_08288.jpg | SRC_154 | train

Novelty 0.557379 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

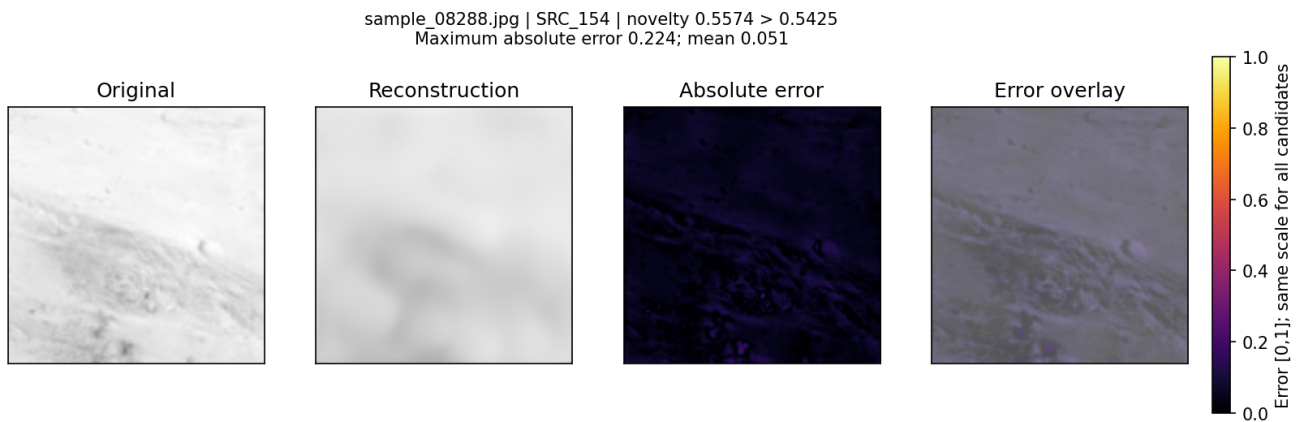


Figure 8. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. A pale smooth upper region meets a darker, rougher diagonal band in the lower half. Small rounded marks and local texture remain visible. Reconstruction smooths the band, and the larger errors follow the rougher lower region.

Possible physical explanation. A contact between a fine bright surface covering and rougher exposed terrain could explain the contrast. Wind redistribution of bright dust is one plausible surface process, without requiring unusual material.

Alternative and limit. Illumination across relief or image contrast processing could produce the same contrast. The isolated crop provides no reliable slope direction, physical scale or compositional evidence. Its high latent novelty does not establish a geological anomaly.

Supplied metadata: latitude 0.0, longitude 180.0, sun angle 54.770875, season N-summer, resolution 0.5.

Process background, not crop identification: <https://www.jpl.nasa.gov/images/pia03789-cerberus-wind-streaks/>.

Phase 3 Candidate 3

sample_04609.jpg | SRC_131 | train

Novelty 0.556367 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

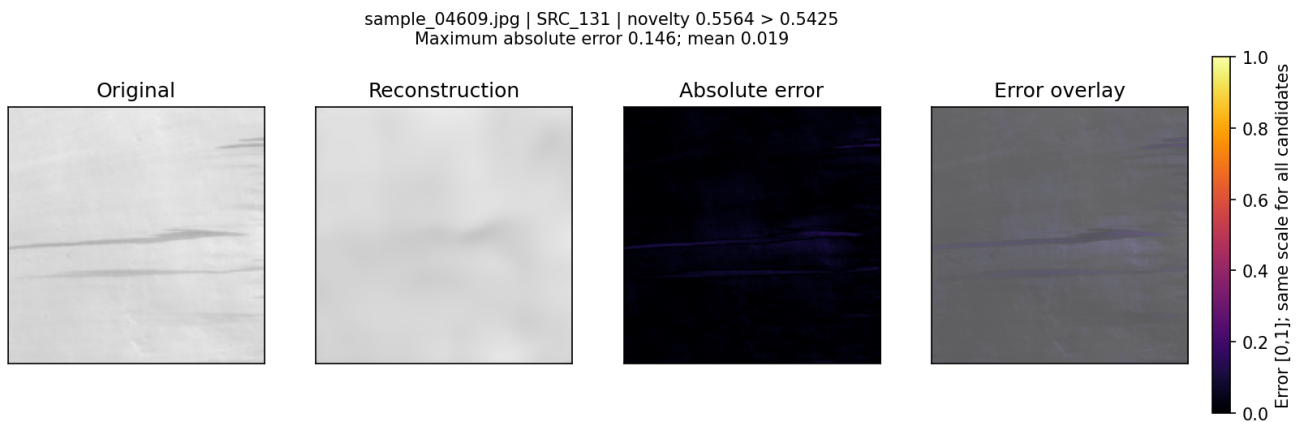


Figure 9. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. Several thin, nearly parallel dark streaks cross a bright, relatively smooth background; two long streaks dominate the middle. The decoder blurs the narrow lines, so the error map traces them even though average reconstruction error is low.

Possible physical explanation. Wind-related removal or redistribution of a bright dust coating could expose darker streaks. This is an ordinary Martian process that can create strong directional contrast.

Alternative and limit. Shadows, image striping or another acquisition artifact remain possible. Orientation alone cannot distinguish a wind streak from these alternatives, and neither wind direction nor slope can be recovered reliably from this crop.

Supplied metadata: latitude 10.0, longitude 180.0, sun angle 60.540464, season N-autumn, resolution 0.5.

Process background, not crop identification: <https://www.jpl.nasa.gov/images/pia07132-wind-streaks-among-flows/>.

Phase 3 Candidate 4

sample_00840.jpg | SRC_154 | train

Novelty 0.555589 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

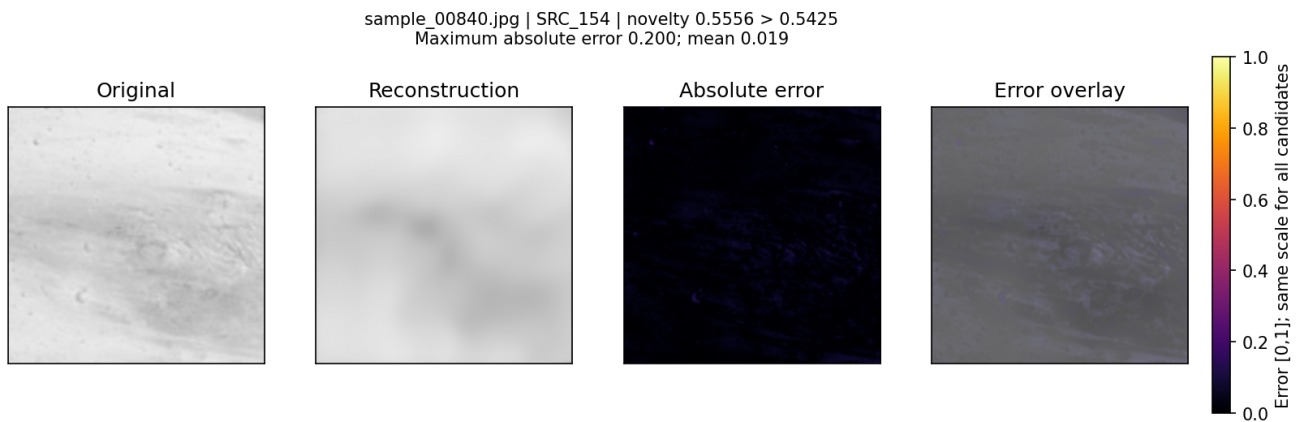


Figure 10. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. A bright background contains a broad diagonal textured band and scattered small rounded depressions or marks. Reconstruction preserves the diffuse band but removes many small details. Error follows the textured band and local marks.

Possible physical explanation. Uneven bright dust cover over a rougher surface could produce this combination of smooth and textured patches. The marks may include small depressions, but their origin cannot be resolved here.

Alternative and limit. Shading or contrast processing could mimic material differences. This crop shares source SRC_154 with two other selected examples, so their similarity may reflect one acquisition rather than independent unusual geology.

Supplied metadata: latitude 0.0, longitude 180.0, sun angle 54.770875, season N-summer, resolution 0.5.

Process background, not crop identification: <https://www.jpl.nasa.gov/images/pia03789-cerberus-wind-streaks/>.

Phase 3 Candidate 5

sample_05391.jpg | SRC_154 | train

Novelty 0.555249 exceeds 0.542471. Conditional threshold-bootstrap flag frequency is 100.0%, not a posterior anomaly probability.

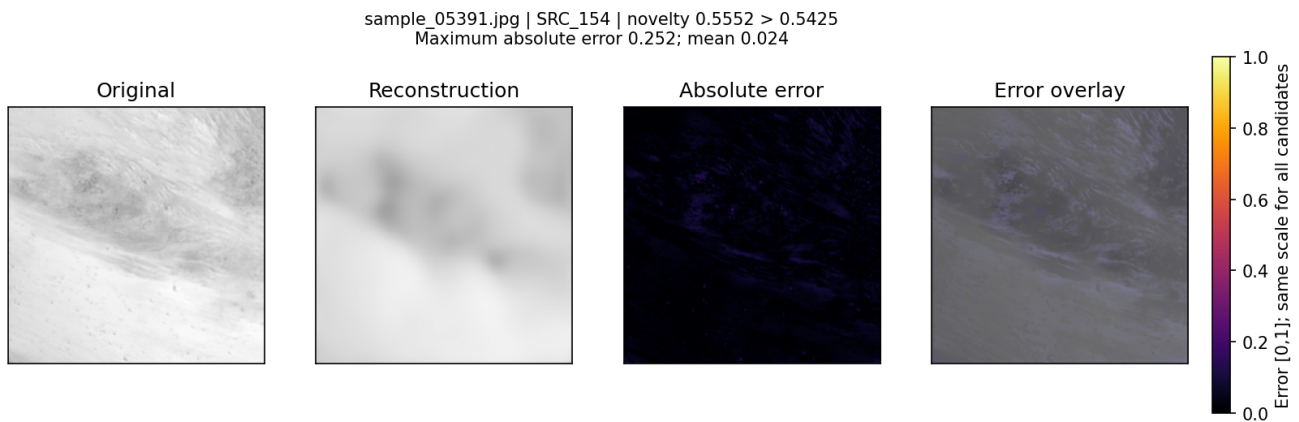


Figure 11. Input, reconstruction, absolute error and discrepancy overlay. Standardized runs additionally show the raw crop. Error refers to the model-input domain and is not direct attribution for Isolation Forest.

What I can see. A rough, striated upper region borders a brighter, smoother lower area along an oblique boundary. Fine linear textures dominate the upper part, and the reconstruction reduces them to broad smooth forms. The error map mainly follows that lost texture.

Possible physical explanation. A transition between a dust-mantled surface and a more exposed textured substrate could explain the boundary. Wind erosion or deposition is a plausible process, but this image alone does not identify the landform.

Alternative and limit. Relief-dependent lighting or radiometric processing remains an alternative. Three of the five selected crops come from SRC_154; source conditions may influence the repeated selection. No injected object is confirmed by this panel.

Supplied metadata: latitude 0.0, longitude 180.0, sun angle 54.770875, season N-summer, resolution 0.5.

Process background, not crop identification: <https://www.jpl.nasa.gov/images/pia03789-cerberus-wind-streaks/>.

Phase 4 My engineering notes

v1: I established a baseline

I trained a 128-dimensional MSE autoencoder from random initialization. Fine texture was smoothed even though validation MSE reached 0.002694. I used that observation to motivate a loss change.

v2 and v3: I changed one factor at a time

Adding SSIM raised validation SSIM from 0.6611 to 0.6717, with a small MSE cost. Increasing the latent vector to 256 then gave MSE 0.002599 and SSIM 0.6738. That is a reconstruction gain, not a hidden-label accuracy result.

I corrected the threshold and checked repetition

The adjusted-boxplot cutoff exceeded every v1 score. I retained that failure, inspected the multimodal distribution, and adopted the fixed mixture-envelope rule with whole-source bootstrap uncertainty. A controlled increase from 400 to 2,000 trees improved forest repeatability. Independent training then exposed fragile candidate membership and a bright-image concentration.

v4: I tested brightness normalization

I standardized each crop without fitting dataset-level statistics. The controlled brightness response improved by 91.4% at the 95th percentile, and the median brightness percentile of the five selected crops fell from 99.90 to 47.51. Forest repeatability weakened and all five contained black strips or borders. I recorded both outcomes.

v5: I tested an exact-zero footprint heuristic

I excluded border-connected exact zeros from normalization statistics and filled them with 0.5. The controlled brightness response improved by 99.87% relative to v3, but minimum forest Jaccard fell to 0.227. Near-black edge fragments remained. I did not silently broaden the mask after seeing selected examples.

I repeated the stronger standardized variant

For v4, changing initialization gives flag-set Jaccard 0.207 and full-ranking correlation 0.867; changing the source split gives 0.360 and 0.866. The three contrast runs flag 24, 11 and 10 crops, with 4 in their intersection. On 1,253 crops held out from both source partitions, rank correlation is 0.897. For raw v3, the corresponding initialization/split Jaccards remain 0.150/0.294, with zero crops flagged in all three raw runs. These are limited sensitivity checks, not detection accuracy. Initialization repeats change both encoder and forest seeds; separate forest-only repeats isolate detector randomness.

I recovered interrupted runs and retained the evidence

Concurrent processes encountered a paging-file error and temporary-checkpoint write failures. I verified the existing checkpoints, removed only the failed temporary files and resumed with unchanged configurations. I then ran GPU work sequentially and used reversible filesystem compression to recover space. The saved execution is local, although the notebook is prepared for Colab.

My final decision

I retain v3 as the screening reference. v4 and v5 reduce a specified brightness nuisance, but their minimum forest-seed flag overlaps fall from 0.667 for v3 to 0.467 and 0.227. All five leading v4 examples contain conspicuous black strips or borders, and the v5 exact-zero mask leaves near-black edge fragments. That evidence does not support replacing the reference with a more reliable geological detector. v3 itself remains brightness-confounded and unstable across independent training runs; I do not present its candidates as confirmed anomalies.

References and reproducibility

Competition inputs. DA PS 26'.docx, DATA_DESCRIPTION.docx, RULES AND REGULATIONS.docx and the supplied archive.

SSIM. <https://www.cns.nyu.edu/~lcv/ssim/>

Resize-convolution. <https://distill.pub/2016/deconv-checkerboard/>

Isolation Forest. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html>

Gaussian mixtures. https://scikit-learn.org/stable/auto_examples/mixture/plot_gmm_selection.html

Connected-component labelling.

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.label.html>

The readable Colab notebook contains the preprocessing, model, training and scoring code directly. Completed configurations, checkpoints, manifests, per-epoch histories and numerical tables accompany the analysis. A fresh runtime needs the supplied private dataset ZIP. No pretrained weights or outside training images are used.

The recorded execution was local. Epoch wall times include interruptions and are not controlled hardware benchmarks. The independent checks are a limited sample of seeds and source assignments. Hidden labels would be needed to establish actual detection performance.

Private repository access, organizer collaborators and team details remain competition logistics. Local packaging does not submit the project.